

Non-asymptotic confidence bands on the probability an individual benefits from treatment (PIBT)

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Introduction

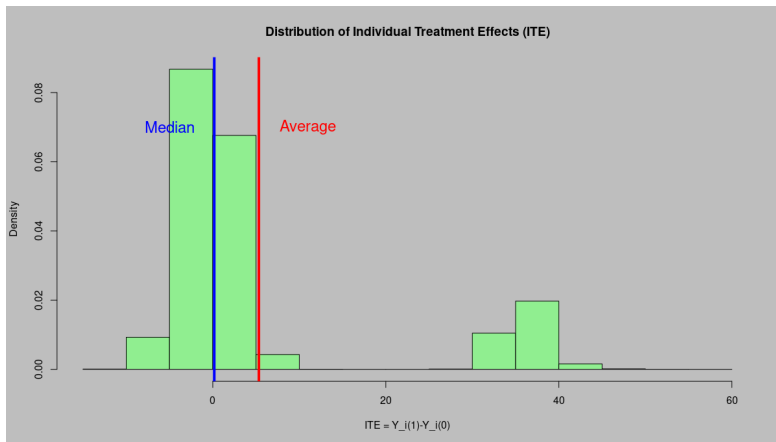


Figure: The mean as a summary of the individual treatment effects.

Introduction

In Ruiz and Padilla (2022), we seek **closed-form**, non-asymptotic confidence bands on the **inestimable** probability an individual benefits from treatment (**PIBT**):

- ① $\Pr(Y_i(1) - Y_i(0) > \delta)$
- ② $\Pr(Y_i(1)/Y_i(0) > \delta)$ (positive outcomes)
- ③ $\Pr(Y_i(1) = 1, Y_i(0) = 0)$ (binary outcomes)
- ④ $\Pr(Y_i(1) - Y_i(0) > \delta | X_i = x)$.

(Can instead call these probability individual is harmed.)

The challenge

WLOG, consider:

$$\Pr(Y_i(1) - Y_i(0) > \delta).$$

- ① Cannot be estimated.
- ② Not even with matching.

The challenge

Probability an individual benefits from treatment:

$$\theta^L(\delta) \leq \Pr(Y_i(1) - Y_i(0) > \delta) \leq \theta^U(\delta).$$

- ① We have to bound this quantity, then estimate those bounds.
- ② Two forms of uncertainty: The best possible bounds, and **our estimation error for these bounds.**

- Extension of existing **plug-in estimation** framework (Fan and Park, 2010).
- **Distribution-free** bounds on CDF of difference (any possible dependence). (Williamson and Downs, 1990; Makarov, 1982).
- Non-parametric Empirical CDF deviation inequality (Dvoretzky et al., 1956; Massart, 1990).
- Neyman-Rubin causal model; marginal and conditional ignorability (Imbens and Rubin, 2015).

Insight 1: Practical Interpretation

With $(1 - \alpha) \times 100\%$ confidence, PIBT is contained between:

[Estimator for lower bound on PIBT] – [Margin of error for PIBT bounds]

and

[Estimator for upper bound on PIBT] + [Margin of error for PIBT bounds].

Insight 2: Power Analysis For Randomized Controlled Trial

- Margin of error for PIBT bounds = $\left(\frac{\log(4/\alpha)}{2}\right)^{\frac{1}{2}} \left(n_0^{-\frac{1}{2}} + n_1^{-\frac{1}{2}}\right)$ with $(1 - \alpha) \times 100\%$ confidence.

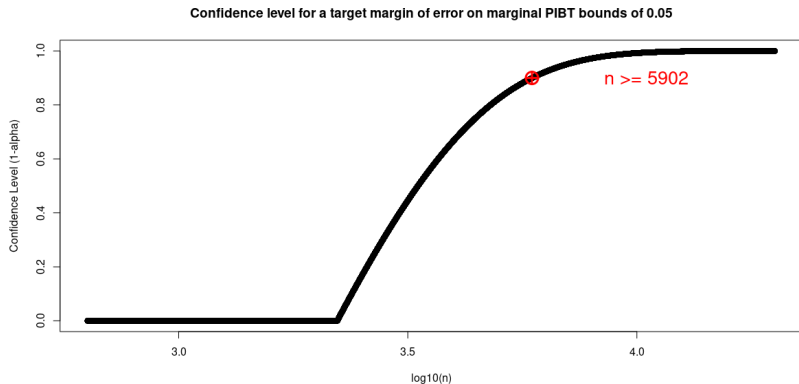


Figure: Power analysis based on our *Theorem 1* (Ruiz and Padilla, 2022). Here, $n_0 = n_1$, but can deal with differing sample sizes.

Insight 3: Heterogeneity in PIBT

- **Simultaneous inference** at all strata x , thresholds δ :

$$\sup_{x, \delta} \left\{ \left| \hat{\theta}^L(\delta, x) - \theta^L(\delta, x) \right| \vee \left| \hat{\theta}^U(\delta, x) - \theta^U(\delta, x) \right| \right\}.$$

- Key: **impute** draws from conditional outcome distribution

$$\hat{\mu}_w(x) + \{Y_i(w) - \hat{\mu}_w(X_i)\} \stackrel{d}{\approx} Y_i(w) | X_i = x.$$

- Target Confidence Level **Correction**:

$$1 - \alpha - \Pr \left(\bigvee_{w=0,1} \sup_x |\hat{\mu}_w(x) - \mu_w(x)| > t_w \right).$$

Insight 3: Heterogeneity in PIBT

At sample size n , what is the uniform Margin of Error on efficacy bounds at the 90% confidence level?

Number of Features $\color{red}\bullet$ $p = 2$ $\color{olive}\bullet$ $p = 5$ $\color{teal}\bullet$ $p = 10$ $\color{blue}\bullet$ $p = 25$ $\color{violet}\bullet$ $p = 50$

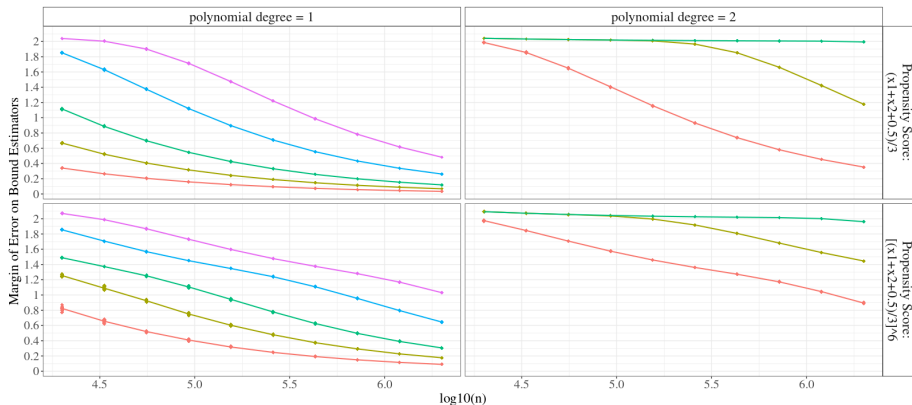


Figure: Power analysis based on our *Theorem 3* applied to canonical linear model (Ruiz and Padilla, 2022).

Conclusion

- We provide closed form high probability concentration inequalities for our estimation of bounds on $\Pr(Y_i(1) - Y_i(0) > \delta)$.
 - i.e. a closed-form “Margin of Error” that allows for a power analysis.
- We extended our theory to case with pre-treatment covariates.
 - A similar power analysis is possible.

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Time to event outcomes?

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